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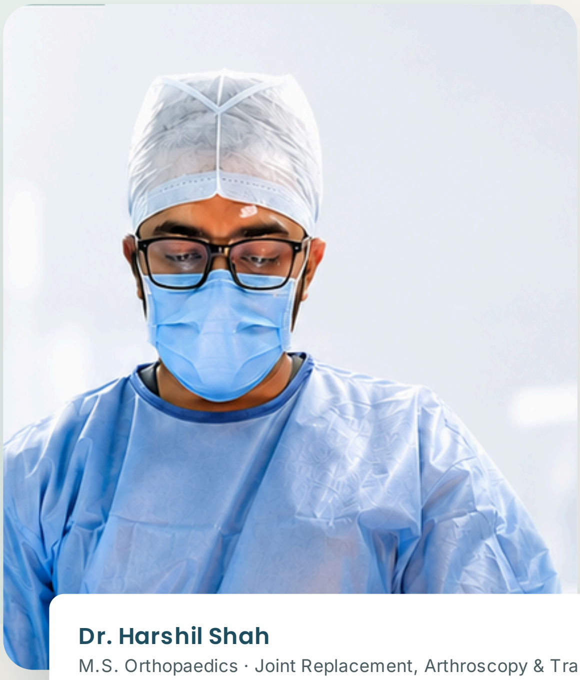
Understand your knee. Choose what's right for you.

Conventional, robotic-assisted & ERAS day-care joint replacement — the questions worth asking, answered plainly.

3 Pathways
Explained side-by-side

Myths, busted
Plain-spoken FAQ

ERAS-based
Faster, safer recovery



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INTRODUCTION

Understand your options. Ask the right questions. Choose what's right for you.

Knee replacement has evolved significantly. Today, successful recovery isn't determined by a single technology — it's the combination of the right patient, the right indication, the right surgical technique, ERAS-based recovery and rehabilitation.

THE MODERN EQUATION

Right Patient + Right Indication + Right Surgical Technique + Appropriate Technology + ERAS-Based Recovery + Early Mobilisation + Structured Physiotherapy + **Safe Discharge**

> *"The goal is not just to replace your knee. The goal is to help you get back to your life."*

Precision matters — but it's one part of a much larger picture.

The right questions matter as much as the right technology.

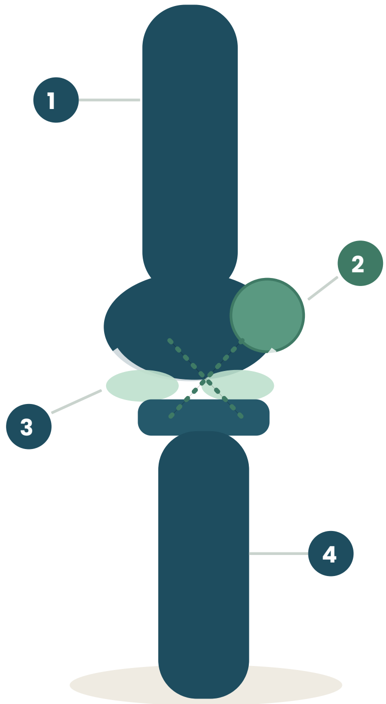
Recovery is a structured journey, not a single decision.

Grounded in patient-education resources from the Hospital for Special Surgery (HSS) and ERAS® Society principles — written for education, not diagnosis.

THE BASICS

What is the knee joint?

One of the largest and most complex joints in the body — connecting the femur (thigh bone), tibia (shin bone) and patella (kneecap).



1. **Femur**
The thigh bone, forming the upper part of the joint — covered in smooth articular cartilage.

2. **Patella**
The kneecap, gliding in a groove at the front of the femur as the knee bends.

3. **Menisci & ligaments**
Crescents of cartilage cushion and distribute load; crossing and side ligaments hold the joint stable.

4. **Tibia**
The shin bone, forming the lower part of the joint and bearing your body weight.

A simple way to understand it — Bone + Cartilage + Meniscus + Ligaments + Muscles = Stability + Movement + Load Bearing.

Articular cartilage covers the joint surfaces, letting the femur and tibia glide with low friction.

Muscles and tendons — especially the quadriceps — drive movement and support the joint.

GETTING STARTED

What is knee arthritis?

Knee arthritis occurs when the protective cartilage and other joint structures progressively deteriorate. This can cause pain, stiffness, swelling and reduced movement — osteoarthritis is the most common form.

COMMON SYMPTOMS

- Pain while walking
- Stair difficulty
- Difficulty rising from a chair
- Stiffness
- Swelling
- Reduced walking distance
- Standing discomfort
- Sleep or daily-life disruption

WHY DOES KNEE ARTHRITIS DEVELOP?

- Increasing age
- Previous knee, ligament or meniscus injury
- Previous knee surgery
- Obesity or excess body weight
- Genetic / familial factors
- Repetitive joint loading
- Inflammatory arthritis & other joint diseases
- Abnormal lower-limb alignment

When should knee replacement be considered? — generally when significant arthritis causes persistent pain or functional limitation despite appropriate non-surgical treatment. The decision should consider symptoms, examination and function — not an X-ray alone.

Source: Hospital for Special Surgery (HSS), patient education resources.

Recognise the pattern of symptoms worth discussing with a surgeon.

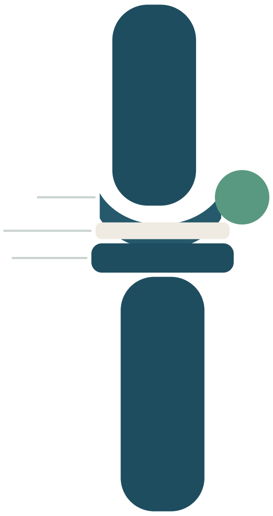
Non-surgical treatment is generally tried first, where appropriate.

Function and examination guide the decision — not an X-ray alone.

UNDERSTANDING THE SURGERY

What is a total knee replacement?

A total knee replacement (TKR/TKA) resurfaces damaged areas of the knee with artificial components. The femur receives a femoral component, the tibia a tibial component, and a polyethylene bearing sits between them. A patellar component may be used when appropriate.



FEMORAL COMPONENT

Metal alloys such as cobalt-chromium or titanium-based materials, depending on implant system.

TIBIAL COMPONENT

Metal baseplate, commonly titanium or cobalt-chromium depending on system.

POLYETHYLENE BEARING & PATELLA

Medical-grade polyethylene provides the bearing surface and, when appropriate, the patellar resurfacing component.

FIXATION

Cemented or cementless, depending on bone quality, patient factors, implant system and surgeon judgement.

How long does it last? — modern knee replacements are designed for long-term durability, but no implant lasts forever. Longevity depends on age, activity, body weight, implant design, surgical technique, bone quality and injury.

MATERIALS AT A GLANCE

- Titanium
- Cobalt-Chromium
- Medical-Grade Polyethylene
- Cemented / Cementless Fixation

MUSCLE-SPARING KNEE REPLACEMENT

What is a subvastus approach?

Surgical exposure is performed beneath the vastus medialis portion of the quadriceps, rather than routinely splitting the quadriceps tendon. The objective is to preserve the extensor mechanism and minimise disruption of important soft tissues.

Preserves the extensor mechanism

Works beneath the quadriceps rather than through it, keeping the muscle-tendon unit intact.

Potentially earlier activation

May support earlier quadriceps activation in the days after surgery.

Supports early mobilisation

Can help patients begin guided walking sooner, as part of an ERAS pathway.

An accelerated rehab pathway

Fits within a structured, accelerated physiotherapy and recovery plan where appropriate.

Why can this be useful? — no single surgical approach is ideal for every patient. The approach should be selected according to anatomy, deformity, previous surgery, implant requirements, surgeon experience and safety.

An important distinction — "muscle-sparing" does not mean "muscle-free surgery." Knee replacement remains a major operation, and appropriate exposure is essential for safe and accurate implant placement.

CONVENTIONAL VS. ROBOTIC KNEE REPLACEMENT

Robotic surgery is a tool — not a replacement for the surgeon

Robotic-assisted knee replacement uses computer-assisted planning and a robotic platform to help the surgeon with patient-specific planning, anatomical assessment, bone preparation, implant positioning and, depending on the system, soft-tissue balancing. The surgeon remains in control.

Feature	Conventional	Robotic-Assisted
Surgeon controls surgery	Yes	Yes
Established technique	Yes	Yes
Patient-specific planning	Yes	Yes
Computer-assisted planning	Not standard	Yes
Real-time tracking	Technique-dependent	Yes
Bone preparation	Surgeon-guided	Robot-assisted
Implant positioning	Surgeon-guided	Computer-assisted
Proven clinical outcomes	Yes	Yes
Automatically better for everyone?	No	No

Both approaches are established, evidence-backed techniques with proven outcomes. Robotic assistance adds computer-guided precision — it doesn't replace the surgeon's judgement, and it isn't automatically the better choice for every patient.

Established techniques, either way

Evidence-based, proven outcomes

Always surgeon-led and controlled

CONVENTIONAL VS. ROBOTIC — CONTINUED

Don't fall for technology-only marketing

A robot cannot replace clinical judgement, surgical experience, correct patient selection, appropriate implant selection, soft-tissue balancing, good anaesthesia, pain management or physiotherapy.

The best technology is the one that is appropriate for the patient.

WHAT A ROBOTIC SYSTEM MAY ASSIST WITH

- Patient-specific planning
- Anatomical assessment
- Bone preparation
- Implant positioning
- Soft-tissue balancing*

*Depending on the system used.

ASK YOUR SURGEON

- Why do I need a knee replacement?
- Why is robotic assistance appropriate for my knee?
- Would conventional surgery also be appropriate?
- What is my rehabilitation plan?
- Am I suitable for ERAS / day-care replacement?
- What surgical approach will be used, and why?

Source: Hospital for Special Surgery (HSS), Robotic Knee Replacement Surgery.

A good answer sounds specific — "because of your anatomy and deformity" is a stronger answer than "because it's more advanced." If the reasoning is only about the technology, ask again.

ERAS & DAY-CARE JOINT REPLACEMENT

Enhanced Recovery After Surgery

ERAS is an evidence-based, multidisciplinary approach designed to improve recovery after surgery. It begins before surgery and continues through discharge and rehabilitation.

THE ERAS PATHWAY

Pre-op optimisation → Patient education → Appropriate anaesthesia → Minimally disruptive technique → Opioid-sparing pain control → Early eating & drinking → Early mobilisation → Early physiotherapy → Safe discharge

What is day-care knee replacement? — a carefully selected patient may undergo knee replacement and return home the same day after achieving predefined clinical recovery and safety criteria. It doesn't mean "no hospital care" — it means appropriate hospital care followed by rapid recovery, safe discharge and continued recovery at home.

Conventional pathway	ERAS / day-care pathway
Hospital stay	Routinely planned
Mobilisation	May be delayed in some pathways
IV therapy	Greater use may occur
Discharge	Based partly on planned stay
Hospital stay	Same-day discharge considered for selected patients
Mobilisation	Early mobilisation
Oral therapy	Early transition where appropriate
Discharge	Based on clinical criteria

ERAS & DAY-CARE — CONTINUED

Who is suitable for day-care?

Several factors are weighed together — no single factor decides eligibility on its own.

- General medical and anaesthetic fitness
- Heart / lung health and other medical conditions
- Age, frailty and functional status
- Patient understanding and motivation
- Suitable home environment and responsible caregiver
- Access to medical assistance if required
- Ability to mobilise safely
- Adequate postoperative pain control

Safety comes first — if the patient does not meet the required discharge criteria, the patient remains in hospital.

A very important clarification — ERAS does NOT mean "no antibiotics" or "no painkillers." Appropriate antibiotic prophylaxis may still be required to reduce infection risk, and pain medication remains an important part of recovery. The ERAS principle is effective multimodal, opioid-sparing pain management.

Source: ERAS® Society, Orthopaedics.

Same-day discharge is possible for well-selected patients.

It isn't the right fit for every patient or every procedure.

Clinical safety criteria decide — not the calendar.

THE FULL PICTURE

Your knee replacement journey

Before surgery, day of surgery, and going home — a structured, checkpoint-based path.

Assessment

Clinical examination and X-rays, with additional investigations where needed.

Optimisation

Blood tests · medical assessment · anaesthesia · medication review · nutrition · blood sugar · anaemia.

Education

Understand your surgery · pain plan · physiotherapy · home prep · discharge plan.

DAY OF SURGERY

Anaesthesia → Conventional or Robotic Surgery → Individualised Pain Management → Recovery → Early Food & Fluids → Early Mobilisation → Physiotherapy

Every step is checkpoint-based — the plan adapts to how you're recovering, not the other way around.

> *“The goal is not just to replace your knee. The goal is to help you get back to your life.”*

YOUR JOURNEY — CONTINUED

Before going home — safety checkpoints

Only once these are cleared does your team confirm you're ready for discharge.

- Stable vital signs
- Adequate pain control
- Able to eat and drink appropriately
- No significant nausea or vomiting
- Able to pass urine
- Safe transfers and safe walking with support
- Physiotherapy assessment completed
- No concerning medical or surgical problem
- Patient & caregiver understand instructions; home support available

Only when the checkpoints are cleared: home. If they are not achieved, continued hospital care is appropriate.

Before you leave, you'll have a written recovery plan, a follow-up date, and a clear physiotherapy schedule to work from at home.

GETTING YOU MOVING

Recovery at home & early physiotherapy

Pain control · Swelling control · Knee movement · Quadriceps activation · Walking · Physiotherapy · Gradual return to activity

01

Ankle pumps

Move the ankle up and down regularly to aid circulation.

02

Quadriceps activation

Tighten the thigh muscle as instructed by your physiotherapist.

03

Heel slides

Gently bend and straighten the knee within the advised range.

04

Straight-leg raise

Begin when adequate quadriceps control is present, and as advised.

05

Knee extension & walking

Work toward comfortable knee straightening, and progress walking with the appropriate aid according to your physiotherapy plan.

Pace matters — progress through these exercises at the rate your physiotherapist advises. Rushing ahead can cause swelling and setbacks; staying consistent, even briefly each day, is what builds lasting range and strength.

FREQUENTLY ASKED QUESTIONS

Myths, answered plainly

Does robotic knee replacement guarantee better results?

No. Robotic technology can assist with planning and precision, but conventional knee replacement also has excellent outcomes.

Does the robot perform the surgery?

No. The surgeon remains in control throughout.

Does robotic surgery guarantee faster recovery?

No. Recovery depends on multiple factors — health, surgical technique, pain management and rehabilitation.

Does muscle-sparing surgery mean no muscle is cut?

No. A subvastus approach aims to preserve the extensor mechanism, but appropriate exposure is still required.

Can everyone have day-care knee replacement?

No. It requires careful patient selection and achievement of postoperative safety criteria.

Does day-care mean no painkillers?

No. The goal is effective pain control with multimodal, opioid-sparing treatment.

Does ERAS mean no antibiotics?

No. Antibiotics may be appropriately used for infection prevention per surgical protocol and patient circumstances.

What happens if I'm not ready to go home?

You stay in hospital. Safety is more important than same-day discharge.

Bring this guide to your consultation — use it to ask the questions on page 08, and to make the most of your time with your surgeon.

- Both conventional and robotic surgery have excellent, proven outcomes.
- Safety criteria — not convenience — decide same-day discharge.
- Recovery is a team effort — surgery, pain control and physiotherapy together.

The modern approach

The goal is not just to replace your knee — it's to help you get back to your life.

Evidence-based patient education drawing on Hospital for Special Surgery (HSS) resources and ERAS® Society principles. For general education — it does not replace an individual medical consultation.

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